

Lab12: Homework

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2026-05-09

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Section 1: Proportion of G|G in a population

We read the file from the provided link Ensemble https://useast.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39895045-39895145;v=rs8067378;vdb=variation;vf=959672880#373531tablePanel.

Q5: What proportion of the Mexican Ancestry in Los Angeles sample population (MXL) are homozygous for the asthma associated SNP (G|G)? HINT: You can filter the displayed genotypes by entering the population code MXL. Then either count those of interest or download a CVS file for this population and use excel or the R functions read.csv(), and table() to answer this question]

```
mxl_data <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl_data)
```

```
## Sample..Male..Female..Unknown.. Genotype..forward.strand.. Population.s. Father
## 1 NA19648 (F) A|A ALL, AMR, MXL -
## 2 NA19649 (M) G|G ALL, AMR, MXL -
## 3 NA19651 (F) A|A ALL, AMR, MXL -
## 4 NA19652 (M) G|G ALL, AMR, MXL -
## 5 NA19654 (F) G|G ALL, AMR, MXL -
## 6 NA19655 (M) A|G ALL, AMR, MXL -
## Mother
## 1 -
## 2 -
## 3 -
## 4 -
## 5 -
## 6 -
```

```
table(mxl_data$Genotype..forward.strand.)
```

```
##
## A|A A|G G|A G|G
## 22 21 12 9
```

```
table(mxl_data$Genotype..forward.strand.) / nrow(mxl_data) * 100
```

```
##
## A|A A|G G|A G|G
## 34.3750 32.8125 18.7500 14.0625
```

So the the G|G percentage for the Mexicans living in the Los Angeles is 14% and this means this genotype percentage is associated to the childhood asthma.

Section 4: Population Scale Analysis

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

Read the data from URL.

```
url <- "https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000172057.6.txt"
data <- read.table(url, header = TRUE)
```

Calculate the sample size (n) for each genotype.

```
sample_size <- table(data$geno)
sample_size
```

```
##
## A/A A/G G/G
## 108 233 121
```

To calculate the median expression of each genotype we Ggroup by genotype and calculate sample size and median

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
results <- data %>%
  group_by(geno) %>%
  summarize(sample_size = n(),
            median_exp = median(exp))
```

```
print(results)
```

```
## # A tibble: 3 x 3
##   geno sample_size median_exp
##   <chr>       <int>       <dbl>
## 1 A/A         108         31.2
## 2 A/G         233         25.1
## 3 G/G         121         20.1
```

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

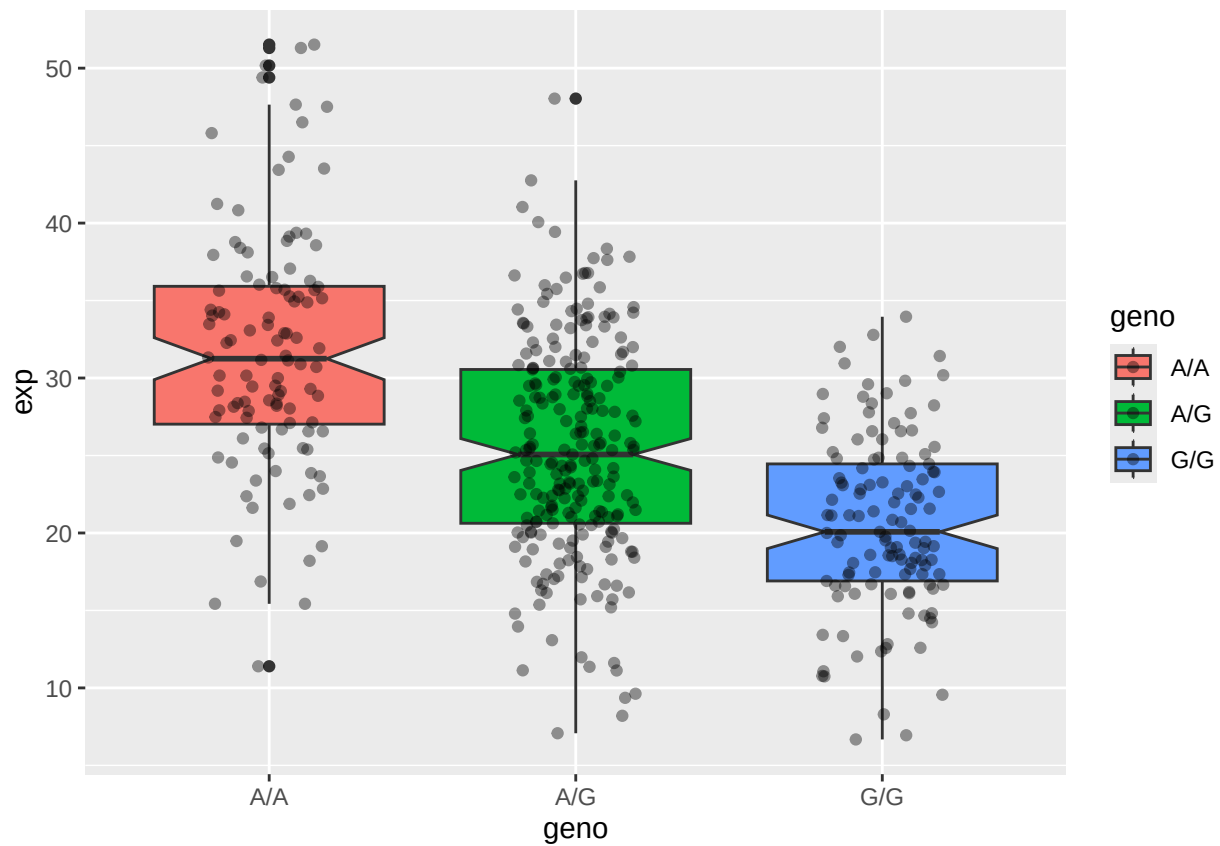
To make the boxplot we need the ggplot.

```
library(ggplot2)
ggplot (data) +
```

```

aes (geno, exp, fill=geno) +
  geom_boxplot (notch=TRUE) +
  geom_jitter(alpha= 0.4, width = 0.2)

```



Yes, the SNP (rs8067378) have a strong effect on the expression of ORM DL3 as distinct separation between the median expression levels of the three genotypes visualized in the boxplot it suggests that this genetic variation is a quantitative trait locus for the ORM DL3 gene. As the number of G alleles increases, the expression of the gene significantly decreases